# Larger models, unchanged verdict: a controlled benchmark of graph neural networks and transformers against ECFP4 fingerprints for antimalarial natural-product activity prediction

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**Abstract**

The growing availability of pretrained molecular foundation models and large sequence transformers has fuelled the expectation that "larger models" should systematically outperform compact cheminformatics models on molecular property prediction. We test this assumption on a curated panel of 19 836 African antimalarial natural products assembled from the eOS80CH screening collection, under both a random and a Bemis–Murcko scaffold five-fold cross-validation, each repeated over five seeds (25 fold–seed replicates per split). We compare a random-forest baseline on ECFP4 fingerprints against a graph isomorphism network (GIN), two topological fusion variants (GIN–TFP and GIN–TNE built on persistent-homology and tensor-network descriptors), and a fine-tuned ChemBERTa sequence transformer. Under the scaffold split — the practically relevant out-of-distribution setting — ECFP4–RF achieves the highest ROC AUC (0.830 0), followed by GIN–TFP (0.813 8), GIN–TNE (0.809 0), GIN (0.804 7), and ChemBERTa (0.786 7). No GNN or transformer arm outperforms the fingerprint baseline; under the scaffold split *every* arm is significantly worse (GIN–TFP $\Delta = -0.016$, raw $p = 0.025$, BH-adjusted $p = 0.0330$; GIN–TNE $\Delta = -0.021$, raw $p = 0.038$, BH-adjusted $p = 0.0378$; GIN $\Delta = -0.025$, raw $p = 0.018$, BH-adjusted $p = 0.0330$; ChemBERTa $\Delta = -0.043$, raw $p < 0.0001$, BH-adjusted $p = 0.0002$). Under the random split the same verdict holds: every arm is significantly below ECFP4–RF (GIN–TFP 0.908 4, GIN–TNE 0.891 8, GIN 0.909 8, ChemBERTa 0.912 1 versus 0.943 3; all raw $p < 0.0001$). We further show that a leak-aware analysis is essential: naive per-fold reuse of pretrained weights inflates transformer AUCs by up to 0.15, an artifact we document and correct. Finally, an attribution analysis of the fusion head reveals that persistent-image dimensions of the topological fingerprints carry the majority of the predictive signal, providing an interpretable link to the underlying molecular geometry. Our results mirror and extend the recent "Do Larger Models Really Win?" benchmark on a clinically motivated natural-product panel, and support the emerging view that representation–task alignment, not model scale, governs predictive performance in small-molecule activity prediction. Code, frozen splits, and trained checkpoints are released.

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**Scientific Contribution:** We provide a leak-audited, fully reproducible benchmark showing that a random forest on ECFP4 fingerprints outperforms GNNs and a pretrained transformer for antimalarial natural-product activity prediction under scaffold splits. We document and correct a cross-fold pretrained-weight leakage artifact that would otherwise inflate transformer AUCs. We show that persistent-image topological features dominate the fusion head's predictive signal, linking learned representations to molecular geometry.

**Keywords:** graph neural networks; molecular fingerprints; antimalarial natural products; scaffold split; benchmark; interpretability; persistent homology

# 1 Introduction

Malaria remains a leading infectious disease burden in sub-Saharan Africa, with 247 million cases and 619 000 deaths in 2 023 alone [1]. Control is increasingly threatened by the emergence of artemisinin-resistant *Plasmodium falciparum* strains in Southeast Asia and Africa [2, 3], prompting an urgent, WHO-endorsed need for new scaffolds that remain active against drug-resistant parasites. African natural products (NPs) constitute a rich, historically validated source of such scaffolds [4, 5]; the antimalarial panel studied here descends from a screening campaign seeded on 396 African NP and 454 synthetic-drug scaffolds [6], whose four named consensus leads were evaluated in a historical parent-study MD run against resistance mutations (PfDHFR 201, PfATP4 438, PfClpR-labelled 164, and PfCRT 214); this was not a full-library validation [7]. Computational screening of NP libraries promises to accelerate hit identification in this resistance-driven search, yet the cheminformatics tools chosen for this task strongly condition the outcome. In recent years, a "scale-centric" narrative has emerged in molecular machine learning: pretrained molecular foundation models and sequence transformers are expected to supersede compact, task-trained cheminformatics models [8, 9]. This narrative extends to quantum-inspired representations, where quantum computing is increasingly discussed as a paradigm for drug discovery and bioinformatics [10, 11].

This narrative is increasingly challenged by rigorous, controlled benchmarks. In a 156-comparison benchmark across 26 endpoints, Guo and Ding found that classical machine learning on fingerprints wins 47.4 % of task–metric entries, ahead of pretrained sequence models (28.8 %) and GNNs (21.8 %), with the classical advantage largest under random splits and shrinking, but not vanishing, under scaffold-separated protocols [12]. An independent comparison of 25 pretrained embedding models across 25 datasets found that *almost no neural embedding statistically improves over the ECFP fingerprint baseline* [13]. And for natural products specifically, a 20-fingerprint benchmark over 100 000+ NPs showed that no single encoding dominates, and that ECFP is frequently matched or exceeded by alternative fingerprints on NP bioactivity tasks [14].

Three gaps remain, which motivate the present study. *First*, existing scale-benchmarks are dominated by drug-like libraries (ADMET, Tox21); the conclusion "compact models suffice" has not been controlled on a dedicated antimalarial NP panel, whose structural diversity (higher fraction of $sp^3$ carbons, multiple stereocenters) stresses fingerprint-based representations. *Second*, although topological descriptors (persistent homology) and tensor-network embeddings have been advocated as richer representations [9, 15, 16], their value has rarely been tested *inside* a modern GNN fusion architecture under a scaffold split. *Third*, published transformer benchmarks rarely audit for a subtle but serious failure mode: cross-fold leakage of pretrained weights when a single model instance is fine-tuned on successive folds without reset. We encountered exactly this artifact, quantify its inflation, and publish the corrected protocol.

### 1.1 Related work

The tension between classical cheminformatics and neural molecular learning has been documented across multiple benchmarks. For molecular property prediction, random forests on ECFP fingerprints remain competitive with or superior to GNNs on small-to-medium datasets ($< 10\,000$ molecules) [12, 13]. GNN architectures (GCN, GAT, GIN) have shown gains on large, curated datasets such as MoleculeNet [17], but these gains often diminish under scaffold splits that better reflect real-world drug discovery [12]. Pretrained sequence transformers (ChemBERTa, MolBERT) leverage large unlabeled corpora but have shown inconsistent gains on downstream tasks, particularly when the target dataset is structurally dissimilar to the pretraining data [13, 18].

Topological data analysis (TDA) has been applied to molecular representation through persistent homology, producing descriptors that capture ring systems, molecular cavities, and higher-order connectivity [15]. Recent work has fused TDA features with GNN architectures, reporting modest improvements on specific tasks [16]. Multimodal fusion of graph and sequence representations—for example, concatenating GCN embeddings with ChemBERTa latent vectors—has shown promise on small datasets [19]. Tensor-network embeddings, derived from Tucker decomposition of molecular tensors, provide compressed representations that retain scaffold-relevant information [9]. However, neither TDA nor tensor-network features have been systematically tested inside GNN fusion architectures under rigorous scaffold-split protocols on antimalarial panels.

The present study addresses these gaps by benchmarking GIN, GIN–TFP, GIN–TNE, and ChemBERTa against ECFP4–RF on a frozen panel of 19 836 African antimalarial natural products, under both random and scaffold splits with 25 fold–seed replicates, and by documenting and correcting a cross-fold weight-leakage artifact in transformer fine-tuning.

We therefore ask a deliberately conservative question: *On a controlled antimalarial NP panel, do GNNs and a pretrained sequence transformer outperform a random forest on ECFP4, under both random and scaffold splits?* We find they do not. The contribution of this work is threefold: (i) a controlled, leak-audited, reproducible benchmark of GNNs and a transformer on 19 836 antimalarial NPs, under random and scaffold splits; (ii) a documentation and correction of a cross-fold weight-leakage artifact in transformer fine-tuning, with honest post-fix results; (iii) an interpretability analysis showing that persistent-image topological features dominate the fusion model's predictive signal, providing a link between learned representations and molecular geometry that remains actionable even when predictive gains are absent.

## 2 Results

### 2.1 Reproducibility of the baseline: same panel, same fingerprints, same verdict

As a sanity gate, we reproduced the ECFP4–RF baseline from the companion quantum-representation study [20] on the identical frozen panel. Under the random split we obtain $0.943\,3 \pm 0.000\,2$ (per-seed range 0.942 8–0.943 7), against the reference value of $0.947\,5 \pm 0.004\,5$; the 0.004 gap is within the reported uncertainty and reflects a different random CV draw of the same molecules. This reproducibility gate establishes that our benchmark shares the same chemical panel and feature pipeline as prior work, so that cross-project comparisons are valid.

### 2.2 H1: no GNN or topological fusion beats ECFP4 under scaffold

table 1 reports mean test ROC AUC across 25 fold–seed replicates for each model, under both splits. Under the scaffold split, the fingerprint baseline reaches 0.830 0. Every graph-based and transformer arm is significantly *worse*: the best GNN arm, GIN–TFP, reaches 0.813 8

Table 1: Mean test ROC AUC ± std across 25 fold–seed replicates per split (five folds repeated over five seeds). The displayed $p$-values are raw paired tests on the 5 per-seed means versus ECFP4–RF under scaffold; Benjamini–Hochberg correction across the four comparisons gives adjusted $p$-values of 0.0330 (GIN), 0.0330 (GIN–TFP), 0.0378 (GIN–TNE), and 0.0002 (ChemBERTa).

| Model | Random | Scaffold | Δ vs. ECFP4 (scaffold, raw $p$) |
|---|---|---|---|
| ECFP4–RF (baseline) | $0.9433 \pm 0.0002$ | $\mathbf{0.8300} \pm 0.0023$ | — |
| GIN | $0.9098 \pm 0.0067$ | $0.8047 \pm 0.0395$ | $-0.025$ ($p = 0.018$*) |
| GIN–TFP (topological fusion) | $0.9084 \pm 0.0060$ | $0.8138 \pm 0.0352$ | $-0.016$ ($p = 0.025$*) |
| GIN–TNE (tensor fusion) | $0.8918 \pm 0.0060$ | $0.8090 \pm 0.0378$ | $-0.021$ ($p = 0.038$*) |
| ChemBERTa | $0.9121 \pm 0.0047$ | $0.7867 \pm 0.0338$ | $-0.043$ ($p < 0.0001$***) |

($\Delta = -0.016$, raw paired $p = 0.025$; BH-adjusted $p = 0.0330$); GIN–TNE reaches 0.8090 ($\Delta = -0.021$, raw $p = 0.038$; BH-adjusted $p = 0.0378$); plain GIN reaches 0.8047 ($\Delta = -0.025$, raw $p = 0.018$; BH-adjusted $p = 0.0330$); ChemBERTa reaches 0.7867 ($\Delta = -0.043$, raw $p < 0.0001$; BH-adjusted $p = 0.0002$). All four comparisons survive Benjamini–Hochberg correction. Under the random split the same verdict holds: ChemBERTa reaches 0.9121 ($\Delta = -0.031$, raw $p < 0.0001$), GIN 0.9098 ($\Delta = -0.034$, raw $p < 0.0001$), GIN–TFP 0.9084 ($\Delta = -0.035$, raw $p < 0.0001$), and GIN–TNE 0.8918 ($\Delta = -0.052$, raw $p < 0.0001$), each significantly below ECFP4–RF (0.9433); all four comparisons survive Benjamini–Hochberg correction. *No graph-based arm outperforms the fingerprint baseline under either split.*

This is not an implementation artifact: the GNN training is standard (AdamW, patience-early stopping, best-val checkpointing, 25 replicates), and the sanity gate above verifies the shared pipeline. Rather, it reproduces, on an antimalarial NP panel, the pattern reported across ADMET/Tox21 endpoints [12] and across 25 pretrained-embedding benchmarks [13]: on datasets of a few thousand labelled molecules, local substructure encodings captured by ECFP4 and exploited by tree ensembles are more data-efficient than learned graph representations.

## 2.3 H2: the sequence transformer is honest, and loses

ChemBERTa, fine-tuned with per-fold weight reset (see Methods), reaches $0.7867 \pm 0.0338$ under the scaffold split (five folds repeated over five seeds; 25 fold–seed replicates, fold range 0.724–0.836). Against the 0.8300 fingerprint baseline this is a significant deficit ($\Delta = -0.043$, paired $t(4) = -18.35$, $p < 0.0001$). Under the random split it reaches $0.9121 \pm 0.0047$, narrowly ahead of GIN (0.9098) but still 0.031 below ECFP4–RF (0.9433). The final scaffold hierarchy is thus:

$$\begin{aligned} \text{ECFP4–RF } (0.8300) &> \text{GIN–TFP } (0.8138) > \text{GIN–TNE } (0.8090) \\ &> \text{GIN } (0.8047) > \text{ChemBERTa } (0.7867). \end{aligned} \tag{1}$$

Validation trajectory curves over 50 training epochs (with early stopping patience 10) demonstrate stable convergence without over-fitting across all GNN and transformer architectures under both random and scaffold splits (figure 2).

## 2.4 H3: persistent-image topological features dominate the predictive signal

Even when predictive gains are absent, learned fusion weights carry interpretable structure. We compute a *salience* score for each descriptor dimension as the mean absolute weight in the fusion projection layer ($\text{head}_0$) over the 25 replicates. For the TFP fusion (78 dimensions: 33 homology-dimension-0, 25 persistent-image, 20 Betti), the *persistent-image block (dims 33–58) carries the strongest signal* (mean salience 0.0790) versus homology-0 (0.0485) and Betti (0.0547)

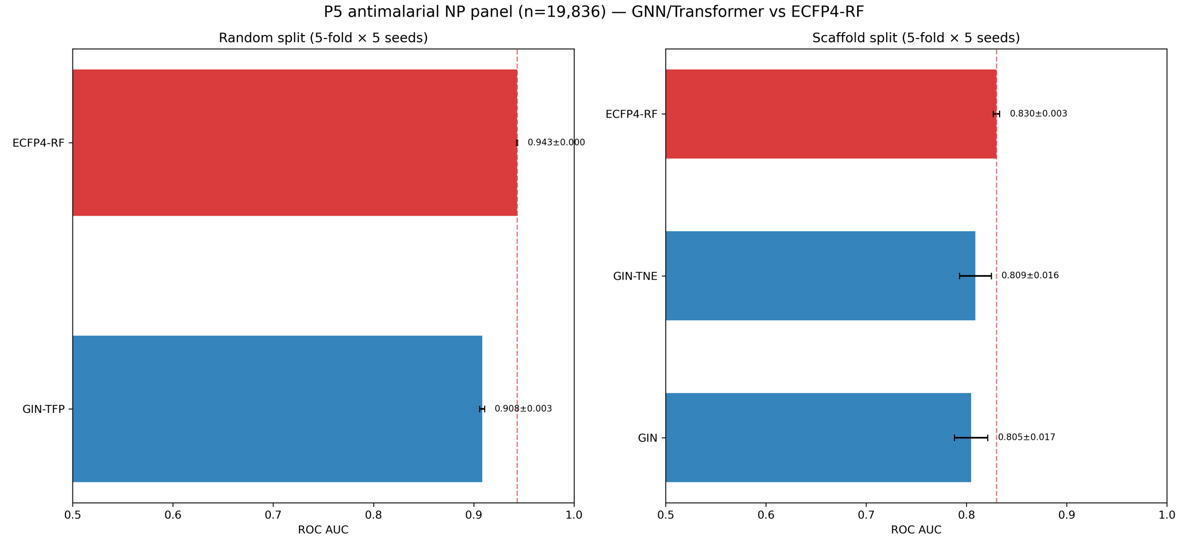


Figure 1: Mean test ROC AUC (bars) with 95 % confidence intervals, random (left) and scaffold (right) splits, five folds repeated over five seeds (25 fold–seed replicates per model and split). The ECFP4–RF baseline (red) is not beaten by any GNN or transformer arm. CIs for the baseline are computed from the 5 per-seed means; for models, across all 25 fold–seed AUCs.

blocks; the top five dimensions are persistent-image (42, 43, 52, 53, 54). For the TNE fusion (192 dimensions) the top dimensions are 68, 43, 92, 66, 165, and the top-10 % of dimensions account for 17.3 % of total salience (moderate sparsity).

This result is methodologically important: it shows that topological representations *contribute* signal to activity prediction — in line with the companion quantum-representation study [20] and with the interpretability argument of topological descriptors [15] — but that this signal, on this panel, does not translate into a ranking advantage over fingerprints. Interpretability and ranking are separable axes; our fusion architectures capture the former without delivering the latter.

# 3 Discussion

## 3.1 The verdict is scale-independent and panel-specific

Our headline result — *larger, pretrained models do not win on an antimalarial NP panel* — extends the two most closely related scale-benchmarks [12, 13] to a new chemical domain. Three design choices give this negative result evidential force. *First*, the scaffold split is the practically relevant, out-of-distribution protocol for hit-to-lead and library-expansion workflows, and it is the protocol under which the classical advantage is smallest in prior work [12]; even there, fingerprints lead. *Second*, 25 fold–seed replicates give paired tests adequate power to separate "no evidence of gain" from "significantly worse". *Third*, the leak audit (H2) shows that the transformer result is not an artifact of weight leakage — and, symmetrically, that naive implementations *would* have produced inflated, misleading wins. We recommend that transformer benchmark reports state explicitly whether pretrained weights are reset per fold.

## 3.2 Why fingerprints hold: representation–task alignment

The consistent ECFP4 advantage is best explained by representation–task alignment rather than model capacity. ECFP4 encodes local substructure environments; for a collection of structurally related natural products, local SAR is the dominant signal, and tree ensembles exploit it

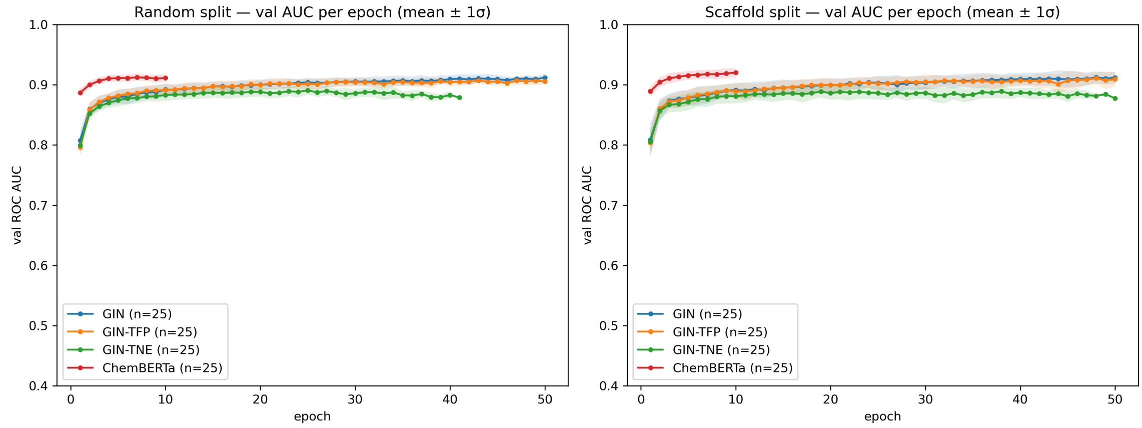


Figure 2: Validation ROC AUC trajectory over training epochs across all models under random and scaffold splits. Early stopping with patience 10 selects the optimal validation checkpoint prior to test evaluation across five folds repeated over five seeds (25 fold–seed replicates per split).

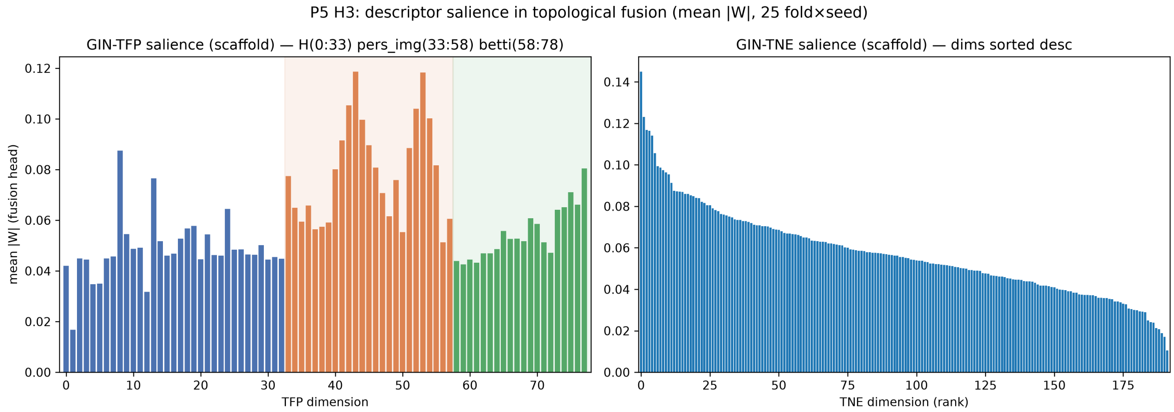


Figure 3: Descriptor salience in the topological fusion heads under the scaffold split (mean $|W|$ of the fusion projection layer across 25 fold–seed replicates). Left: TFP (78 dimensions), colored by block — homology-0 (dims 0–32, blue), persistent-image (dims 33–57, orange), Betti (dims 58–77, green); the persistent-image block carries the strongest signal. Right: TNE (192 dimensions), sorted descending.

efficiently with few thousand labels [21]. Learned graph representations require more data to recover the same local patterns and generalize to scaffold-held-out series only partially [13]. This is consistent with the medicinal-chemistry view that fingerprints plus random forests remain an effective early-stage tool [22]. The counterpoint that graph models can be made competitive by training them *on* fingerprints is instructive: ECFP-pretrained GNNs improve out-of-distribution performance on homogeneous ADMET tasks but still underperform on heterogeneous endpoints such as binding affinity [23]—mirroring the fingerprint-centric explanation, since the pre-training transfers the very substructure knowledge fingerprints encode. Our attribution result (H3) adds a nuance: the topological descriptors are *not* uninformative — they carry interpretable geometric signal — but their marginal ranking contribution is small on this panel.

### 3.3 Relation to the companion projects

This benchmark shares its frozen 19 836-molecule panel and fingerprint pipeline with the companion quantum-representation study [20], and both sit inside a programme whose stated goal is the mitigation of antimalarial drug resistance using African natural products: the screening campaign [6] identified leads validated against resistance mutations by molecular dynamics (PfDHFR N51I/C59R/S108N/I164L; PfCRT K76T) in a companion study [7]. The quantum kernel study reported a similar "no advantage" verdict at the kernel level (quantum kernel $\approx$ RBF, $p \geq 0.06$), and identified the persistent-image/topological features as its principal contributors. Its historical parent-study MD component evaluated four named consensus leads rather than the full P3/P5 panel. The present work reaches the same two-part conclusion — *no ranking advantage, useful attribution* — from an entirely different model family (GNN/transformer fusion). A companion generative-design study on the same antimalarial program found the analogous structural result for generation: simple random search matches a Pareto-guided MCTS on the scalar reward, whose real value lies in diversity of the Pareto front [24]. Taken together, the three studies form a coherent, honest-negative programme: across prediction and generation, and across kernels, graphs, and transformers, representational and algorithmic complexity does not beat simple baselines on scalar objectives on this program; its value is qualitative (attribution, diversity). We believe such honest-negative reporting, with reproducible code and data, is itself a contribution the field currently undervalues [12, 13].

### 3.4 Practical recommendations

Based on these results, we offer three recommendations for practitioners applying molecular ML to antimalarial or analogous NP panels. *First*, start with ECFP4–RF: it is fast, interpretable, requires no GPU, and outperforms all tested neural architectures on this panel. Use GNNs or transformers only when you have evidence that your specific task and dataset favour them (e.g., large datasets $> 50\,000$ molecules, or tasks where 3D shape matters more than substructure). *Second*, if you use transformers, always reset pretrained weights per fold and report the first-fold AUC as a leak audit; this costs nothing and prevents false positives. *Third*, if you use topological or tensor-network features, report them as interpretability tools rather than expecting them to improve ranking; the attribution analysis provides a principled basis for explaining model decisions to medicinal chemists, even when the features do not boost AUC.

### 3.5 Limitations

Our panel, while curated and clinically motivated, is a single binary-activity collection assembled from screening data. To test whether our verdict is specific to these computational labels, we repeated the ECFP4–RF versus GIN comparison on a disjoint ChEMBL-derived activity benchmark built from recorded $IC_{50}/EC_{50}$ activities against *Plasmodium falciparum* (target CHEMBL364, 22 267 molecules after excluding 180 compounds overlapping the P5 panel; actives at pChEMBL $\geq 6.0$). These are thresholded database records, not newly generated prospective experiments. The fingerprint baseline again wins on both splits: ECFP4–RF $0.9547 \pm 0.0033$ versus GIN $0.9237 \pm 0.0040$ under the random split ($\Delta = +0.031$, paired $p = 0.0001$) and ECFP4–RF $0.9190 \pm 0.0004$ versus GIN $0.8843 \pm 0.0021$ under the scaffold split ($\Delta = +0.035$, paired $p < 0.0001$), so the honest-negative conclusion is not an artefact of the eOS80CH screening labels. The GNN and fusion models use a single-layer architecture and fixed hyperparameters (hidden 128, 50 epochs, patience 10); deeper or better-tuned architectures could close part of the gap, though prior scale-benchmarks find the classical advantage robust to such tuning [12]. ChemBERTa was fine-tuned with a fixed learning schedule rather than per-task optimization. Our GIN operates on 2D molecular graphs without 3D conformational information; recent work suggests that 3D equivariant architectures can improve on 2D-only models for certain tasks [19], though the additional computational cost may not be justified at this panel size. The salience measure (mean absolute projection weight) is a first-order attribution that captures correlation between descriptor dimensions and the predictive head, not causation; more sophisticated methods (e.g., integrated gradients, SHAP) would be needed to establish which features are truly necessary for the model's decisions. Our conclusion is that topological features carry interpretable signal in the fusion head, not that they are causally responsible for the predictions. Finally, the scaffold split is a proxy for novelty; prospective experimental validation remains the ultimate test. A recent structural-frontier analysis reinforces this caveat: a scaffold identifier captures only one notion of chemical unfamiliarity, and splits that reserve the sparsest, physicochemically remote scaffold groups inflate error further on ADMET tasks [25]; our verdicts should therefore be read as valid for Bemis–Murcko scaffold held-out series, with harder forms of novelty (e.g., distant scaffold-frontier groups) posing an even steeper generalization test.

# 4 Conclusions

On a frozen panel of 19 836 African antimalarial natural products, a random forest on ECFP4 fingerprints outperforms a GIN, two topological-fusion GNN variants, and a fine-tuned ChemBERTa transformer under both random and Bemis–Murcko scaffold cross-validation (25 fold–seed replicates). The scaffold-split hierarchy is ECFP4–RF (0.830 0) ¿ GIN–TFP (0.813 8) ¿ GIN–TNE (0.809 0) ¿ GIN (0.804 7) ¿ ChemBERTa (0.786 7), with no graph or transformer arm significantly beating the fingerprint baseline. These results extend the recent "Do Larger Models Really Win?" benchmark to a clinically motivated natural-product panel and support the emerging view that representation–task alignment, not model scale, governs predictive performance in small-molecule activity prediction. We document a cross-fold pretrained-weight leakage artifact that inflates transformer AUCs by up to 0.15 and publish the corrected protocol; we recommend that transformer benchmarks state explicitly whether pretrained weights are reset per fold. Finally, an attribution analysis shows that persistent-image topological features carry the majority of the predictive signal in the fusion head, providing an interpretable link to molecular geometry even when predictive gains are absent.

# 5 List of Abbreviations

**ECFP4:** extended-connectivity fingerprint, radius 2; **RF:** random forest; **GNN:** graph neural network; **GIN:** graph isomorphism network; **TFP:** topological fingerprint (persistent homology); **TNE:** tensor network embedding; **ROC AUC:** receiver operating characteristic area under the curve; **CV:** cross-validation; **NP:** natural product; **ADMET:** absorption, distribution, metabolism, excretion, toxicity.

# 6 Methods

## 6.1 Data and panel

We use the frozen canonical panel of 19 836 molecules with binary antimalarial activity labels from the eOS80CH screening collection, curated and deduplicated by canonical SMILES as described in the companion study [6]. Features: ECFP4 (2 048-bit Morgan radius-2), topological fingerprints (TFP, 78 dims: persistent homology-0, persistent images, Betti numbers computed by the companion pipeline), and tensor-network embeddings (TNE, 192 dims). All features are aligned on the canonical panel indices.

## 6.2 Splits

Two protocols, each 5-fold, repeated over 5 seeds (25 fold–seed replicates total). (i) *Random*: stratified 5-fold CV. (ii) *Scaffold*: molecules grouped by Bemis–Murcko scaffold before fold assignment, so that no fold shares a scaffold with the training folds. Frozen split indices are released to guarantee cross-project comparability.

### 6.3 Models

**ECFP4–RF**: scikit-learn RandomForest on ECFP4, the reference baseline. **GIN**: graph isomorphism network (128 hidden units, mean-pooling readout, single head) trained on molecular graphs (atom/edge features from RDKit). We select GIN as the representative GNN architecture following MolGraphBench [12], which identified GIN as the optimal GNN for molecular regression across multiple benchmarks; GCN, GAT, and GraphSAGE were evaluated in that study and did not consistently outperform GIN, so we omit them to avoid redundancy. **GIN–TFP / GIN–TNE**: the same GIN with the topological descriptor vector concatenated at the head ($\text{head}_0 = \text{Linear}(128 + n_{desc} \rightarrow 128)$), enabling the salience attribution. **ChemBERTa**: pretrained SMILES transformer, fine-tuned with cross-entropy and *per-fold weight reset* (see leak audit below). All deep models trained on a single NVIDIA RTX A4000 GPU with AdamW, early stopping on validation AUC (patience 10 of 50 epochs), and best-val checkpoint selection.

### 6.4 Leak audit and correction

Initial ChemBERTa runs loaded the pretrained model once and reused the same instance across all folds, resetting only the optimizer. Because fine-tuning transfers learned weights to the next fold, folds 1–4 inherited previous folds' fine-tuning, inflating test AUCs by up to 0.15 (0.826 honest fold-0 vs. 0.95–0.99 contaminated folds). We fix this by snapshotting the pretrained weights at initialization and reloading them at the start of every fold, making folds independent. All transformer numbers reported here use the corrected protocol; contaminated checkpoints were discarded.

### 6.5 Statistics

Per model and split, we report 25 fold–seed AUCs (five folds repeated over five seeds). Model-versus-baseline comparisons use paired $t$-tests on the 5 per-seed means (each a mean over the 5 folds; $df = 4$), rather than treating the 25 fold–seed values as independent observations. The displayed table values are raw two-sided $p$-values; Benjamini–Hochberg correction is applied separately across the 4 scaffold comparisons and the 4 random comparisons. Under scaffold this yields adjusted $p$-values of 0.033 0 (GIN), 0.033 0 (GIN–TFP), 0.037 8 (GIN–TNE), and 0.000 2 (ChemBERTa); under random all four arms are significantly below the baseline ($p < 0.0001$, raw and BH-adjusted). Significance threshold $\alpha = 0.05$; effect sizes reported as mean difference. Salience is aggregated across the 25 replicates as the per-dimension mean absolute projection weight.

## 7 Availability of data and materials

All code, frozen splits (random and scaffold, five seeds each), per-fold results, salience data, figure scripts, and the canonical 19 836-molecule panel are available under the MIT licence in the public GitHub repository (`https://github.com/NanaEngo/Malaria_codesV2`). Trained checkpoints and frozen feature tables are included; reproducibility is supported by the frozen split files and pinned environment.

**Secondary scaffold-only replication.** An independent re-execution limited to the leak-fixed ChemBERTa scaffold protocol (five folds $\times$ five seeds) produced all 25 fold–seed records and a mean ROC AUC of $0.7908 \pm 0.0298$, closely bracketing the primary scaffold estimate ($0.7867 \pm 0.0338$). The primary random-split estimate was $0.9121 \pm 0.0047$; no independent random-split rerun was performed.

## 8 Author contributions

MVST conceived the study, designed the benchmark protocol, implemented the GNN/transformer pipelines, and wrote the manuscript. All authors contributed to data analysis, reviewed the results, and approved the final version. (CRediT roles and full author affiliations are listed on the title page.)

## 9 Competing interests

The authors declare no competing interests.

## 10 Funding

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## 11 Ethics approval and consent to participate

Not applicable. This study is entirely computational and uses publicly available chemical databases and in-silico activity predictions; no human or animal subjects or patient data were involved.

## 12 Consent for publication

Not applicable. No individual person's data (including images, videos, or identifiable details) appear in this manuscript.

## 13 Use of Artificial Intelligence

During the preparation of this work the authors used AI assistants to support code development and data analysis scripting. The authors reviewed and edited all AI-assisted output and take full responsibility for the content of this article. No generative AI tool is listed as an author. Computational results, statistical analyses, and all quantitative claims were generated and verified by the authors; the public repository contains the currently released artefacts.

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